

Name:-Sharvari Patil

PRN:-B25ET1180

Q.4) Write a c++ program to cube a complex number using a class.

Program:- #include <iostream>

using namespace std;

class Complex

{

float real, imag;

public:

void getData()

{

cout << "Enter real part: ";

cin >> real;

cout << "Enter imaginary part: ";

cin >> imag;

}

void cube()

{

float r = real * real * real - 3 * real * imag * imag;

float i = 3 * real * real * imag - imag * imag * imag;

cout << "Cube of complex number = " << r;

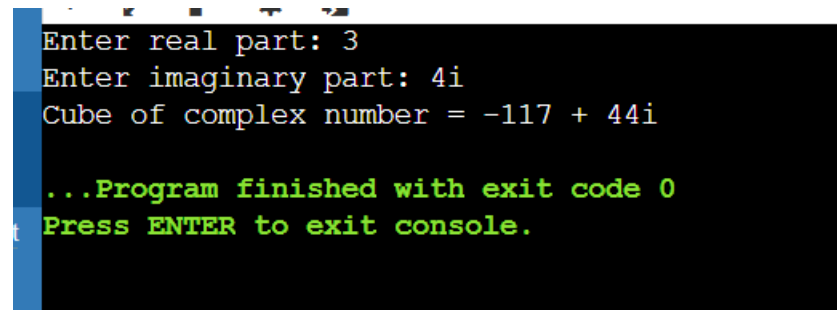
if (i >= 0)

cout << " + " << i << "i";

```
        else  
            cout << " - " << -i << "i";  
    }  
};
```

```
int main()  
{  
    Complex c;  
  
    c.getData();  
    c.cube();  
  
    return 0;  
}
```

Output:

A screenshot of a terminal window with a black background and green text. The text shows the program's execution: it prompts for the real and imaginary parts, calculates the cube of the complex number, and displays the result. It also shows the program finishing with exit code 0 and a prompt to press ENTER to exit the console.

```
Enter real part: 3  
Enter imaginary part: 4i  
Cube of complex number = -117 + 44i  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```